

Total No. of Questions : 4]

SEAT No. :

PD-447

[Total No. of Pages : 2

[6410]-546

T.E. (COMPUTER ENGINEERING) (Insem.)

Statistics and Machine Learning

(HONOURS : Data Science)

(2019 Pattern) (Semester - II) (310503)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Answer Q1 Or Q2 & Q3 Or Q4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) Explain the Inferential Statistics? What are the types? Explain them with examples Difference between Descriptive Statistics & Inferential Statistic.

[7]

b) What is mean by Central tendency? Explain the Concept of

- i) Mean
- ii) Median
- iii) Mode

For each measure with example.

[8]

OR

Q2) a) What is Hypothetical testing, elaborate on the steps of hypothetical testing and the types of Hypothetical testing?

[7]

b) Explain the following tests.

- i) Chi-square test
- ii) T-test

Also explain Bi-variate regression.

[8]

P.T.O.

Q3) a) Explain the Bayes' Theorem with suitable example. Explain Types of Naive Bayes Classifier. Why Naive Bayes Classifiers are useful in Bayes Theorem. [7]

b) What is Z-score, find the Z-score if $x = \{5, 7, 14, 9, 21, 17, 12\}$. [8]

OR

Q4) a) What do you mean by probabilistic models with hidden variable? Explain the different types of probabilistic models with examples. [7]

b) Define Karl Pearson's Coefficient of Correlation and explain its significance. Derive the formula and calculate the correlation coefficient for the following data:

X	10	12	15	18	20
Y	25	30	28	35	40

[8]

